Product Inventory Management System

Team Size:4

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Introduction

The Product Inventory Management System is a backend web application developed using Node.js, Express.js, and MongoDB. The system is designed to help users manage product inventory efficiently through secure APIs.

This application allows users to register an account, log in securely, and perform operations like adding products, viewing inventory, updating product details, and deleting products when necessary.
The system uses JWT (JSON Web Token) authentication to protect private routes and ensure that only authorized users can access the application. Each user can manage only their own products, which ensures privacy and data security.

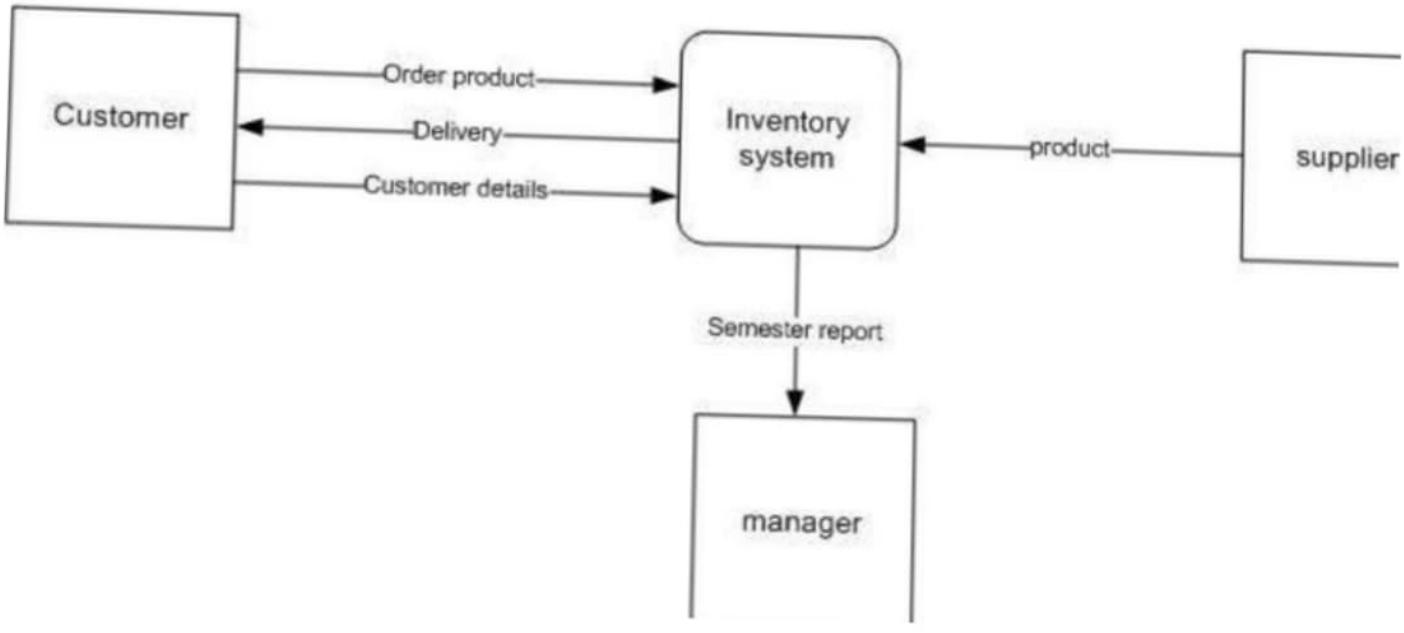
This backend system follows a RESTful API architecture, making it scalable and easy to integrate with any frontend application.
The system is useful for small business owners, students, freelancers, and developers who want to manage product inventory or learn backend API development

Objectives of the System

The system helps businesses maintain accurate records of all products available in stock. It also helps in monitoring the quantity of products and ensures that the business does not run out of stock.

Another objective of the system is to reduce manual work. Manual inventory management requires a lot of time and effort. By using a computerized system, businesses can perform inventory tasks faster and more efficiently.

CONTEXT LEVEL DIAGRAM OF CURRENT SYSTEM

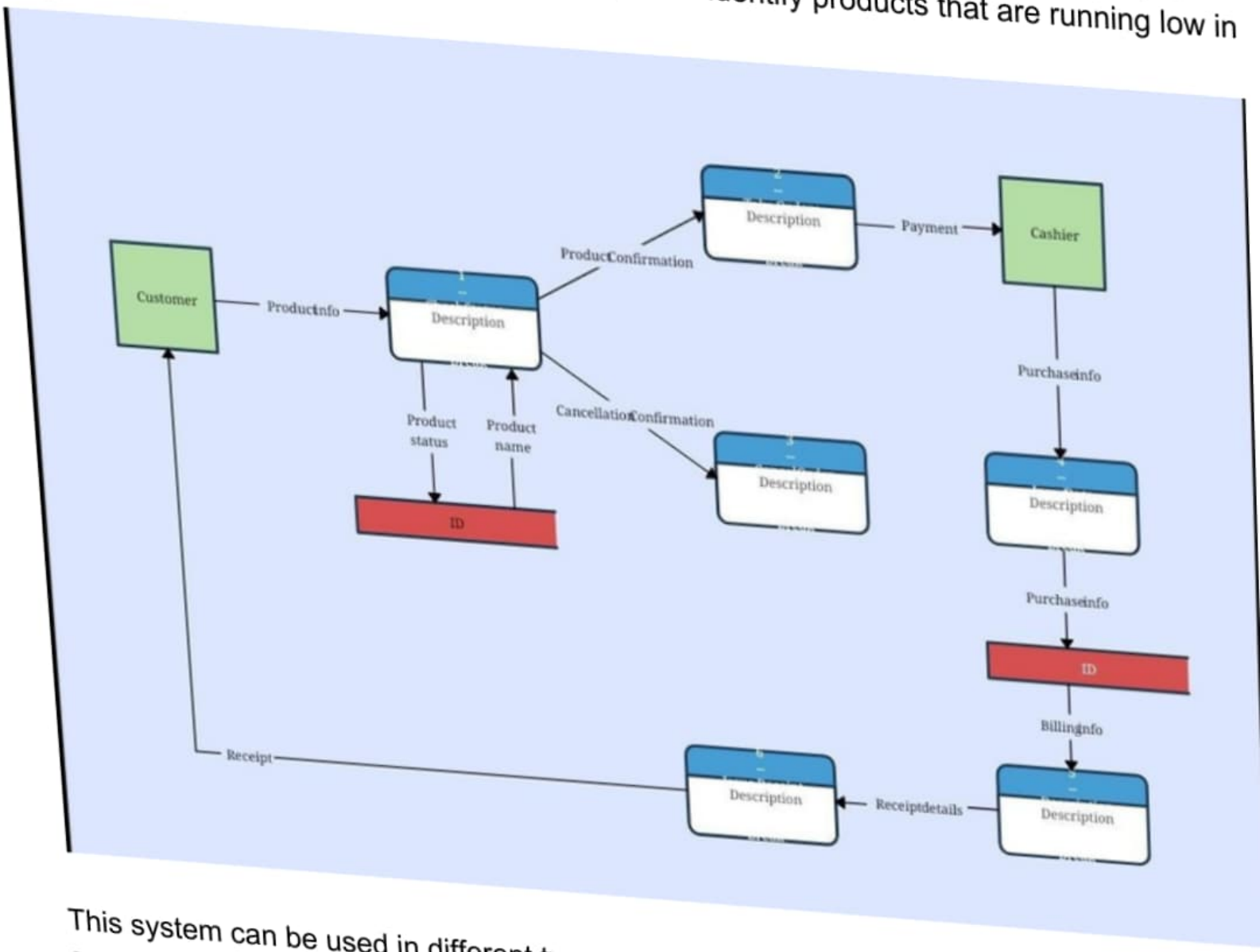


The system also aims to improve data accuracy. When inventory is managed manually, there isa high chance of mistakes. A digital system reduces these errors and ensures reliable data storage.

Scope of the Project

The scope of the Product Inventory Management System includes managing all product-related information in a business. The system stores details such as product name, product ID, quantity available, price, and supplier information.

The system allows users to add new products, update product details, delete products, and view product lists. It also helps track stock levels and identify products that are running low in stock.



This system can be used in different types of businesses such as retail shops, supermarkets, warehouses, and small companies. It helps improve inventory control and ensures smooth business operations.

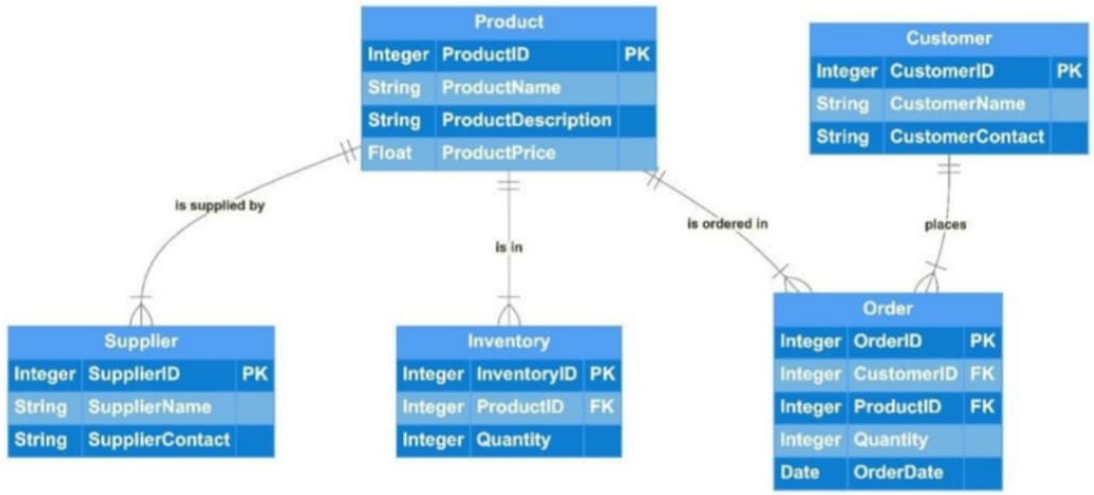
Modules of the System

The Product Inventory Management System is divided into different modules to perform specific tasks. Each module handles a particular function of the system.

Main modules include:

- Product Management Module
- Stock Management Module
- Supplier Management Module
- Report Generation Module

These modules work together to manage inventory efficiently.



Target Users

This system is designed for different types of users:

Small Business Owners

They can use this system to track their product stock and inventory.

Freelancers or Online Sellers

They can store product information such as price, quantity, and description.

Developers

Developers can use this project to learn backend development using Node.js and MongoDB.

Students

Students working on backend or full-stack projects can use this system for learning and practice

Software Requirements

The following software tools are required to develop and run the project.

Node.js

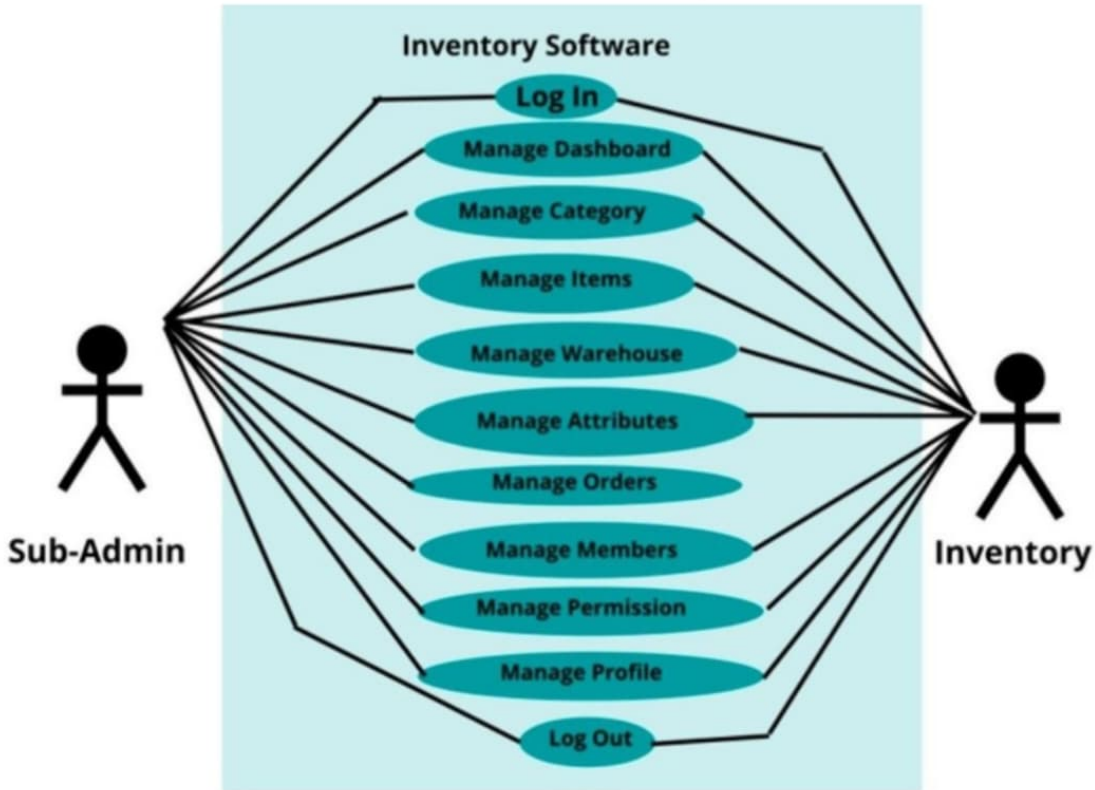
Node.js is used as the JavaScript runtime environment for backend development.

MongoDB

MongoDB is a NoSQL database used to store user and product data.
npm (Node Package Manager)
It helps install and manage project dependencies.

Visual Studio Code

VS Code is used as the development environment for writing and running code.



Knowledge Requirements

Tounderstandand implementthis project,thedeveloper should know:

- JavaScript ES6 concepts
- Node.js and Express.js
- REST API development
- MongoDB database
- JWT authentication
- Middleware usage

These technologies together help in building a secure and scalable backend system.

Database Architecture

The database consists of two collections:

Users Collection

Stores user information such as:

- Name
- Email
- Password (hashed)
- Timestamps

Products Collection

Stores product details such as:

- Product name
- Price
- Quantity
- Description
- User ID reference

Each user can have multiple products, creating a one-to-many relationship between users and products.

Core Features

1. User Authentication

Users can registerandlog inusing their email and password. JWT tokens are generated after login for secure API access.

2. Product Management

Userscanmanagetheirinventoryby adding new products and storing product details.

3. Product CRUD Operations

The system supportsthe followingoperations:
Create – Add new product
Read – View all products
Update – Edit product details
Delete – Remove products

4. Product Search

Userscansearchproductsbased on their name to easily locate items in their inventory.

5. Secure API Access

Allproductroutesareprotectedusing authentication middleware. Only logged-in users can access them.

Project Setup

To run the project, follow these steps:
1.Create the backend project folder.
2.Initialize the Node.js project using:

```
npm init -y
```

3.Install required packages:
npm install express mongoose cors dotenv jsonwebtoken bcryptjs

4.Create an .env file and add:

PORT=5000
MONGO_URI=your_mongodb_connection_string
JWT_SECRET=your_secret_key
5. Connect the MongoDB database.
6.Start the server using:

```
npm start
```

If successful, the console will display:

Server running on PORT
Database Connected

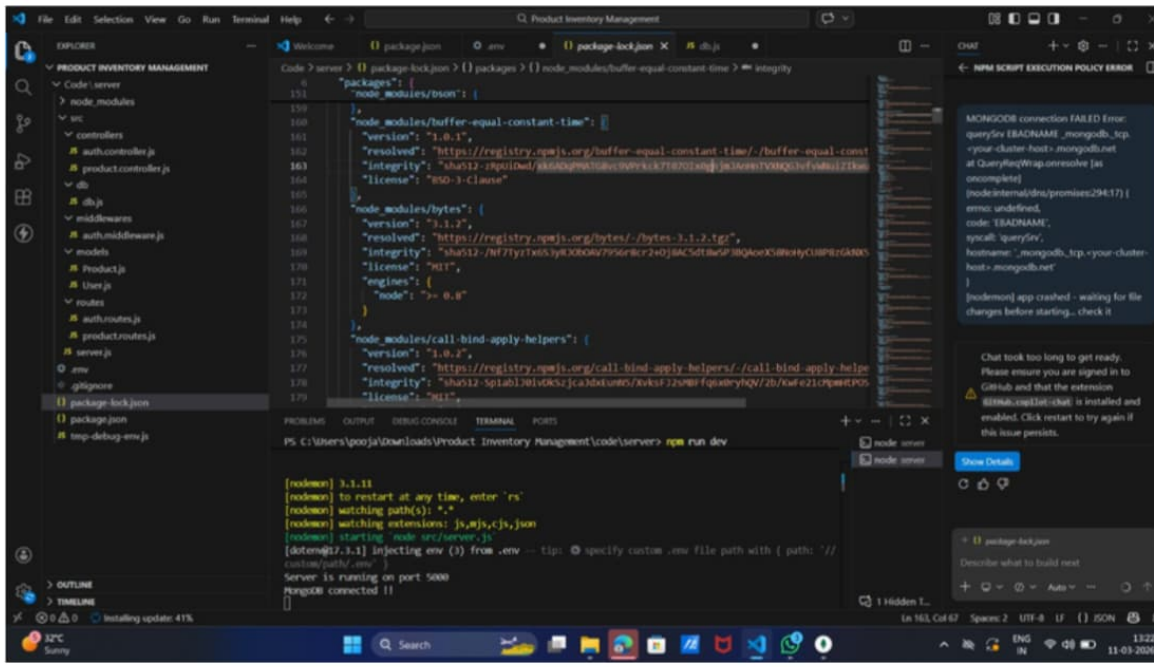
API Testing

The APIs can be tested using Thunder Client in VS Code.

Test the following APIs:
Register User

- Login User
- Create Product
- Get Products
- Update Product
- Delete Product

screenshots



```java
import java.util.Scanner;

public class Inventory {
    public static void main(String[] args)
{

        Scanner sc = new
Scanner(System.in);

        String productName;
        int quantity;
        double price;

        System.out.println("Enter Product
Name:");
        productName = sc.nextLine();

        System.out.println("Enter
Quantity:");
        quantity = sc.nextInt();

        System.out.println("Enter
Price:");
        price = sc.nextDouble();

        System.out.println("\n--- Product
Details ---");
        System.out.println("Product Name:
" + productName);
        System.out.println("Quantity: " +
quantity);
        System.out.println("Price: " +
price);

        sc.close();
    }
}
```

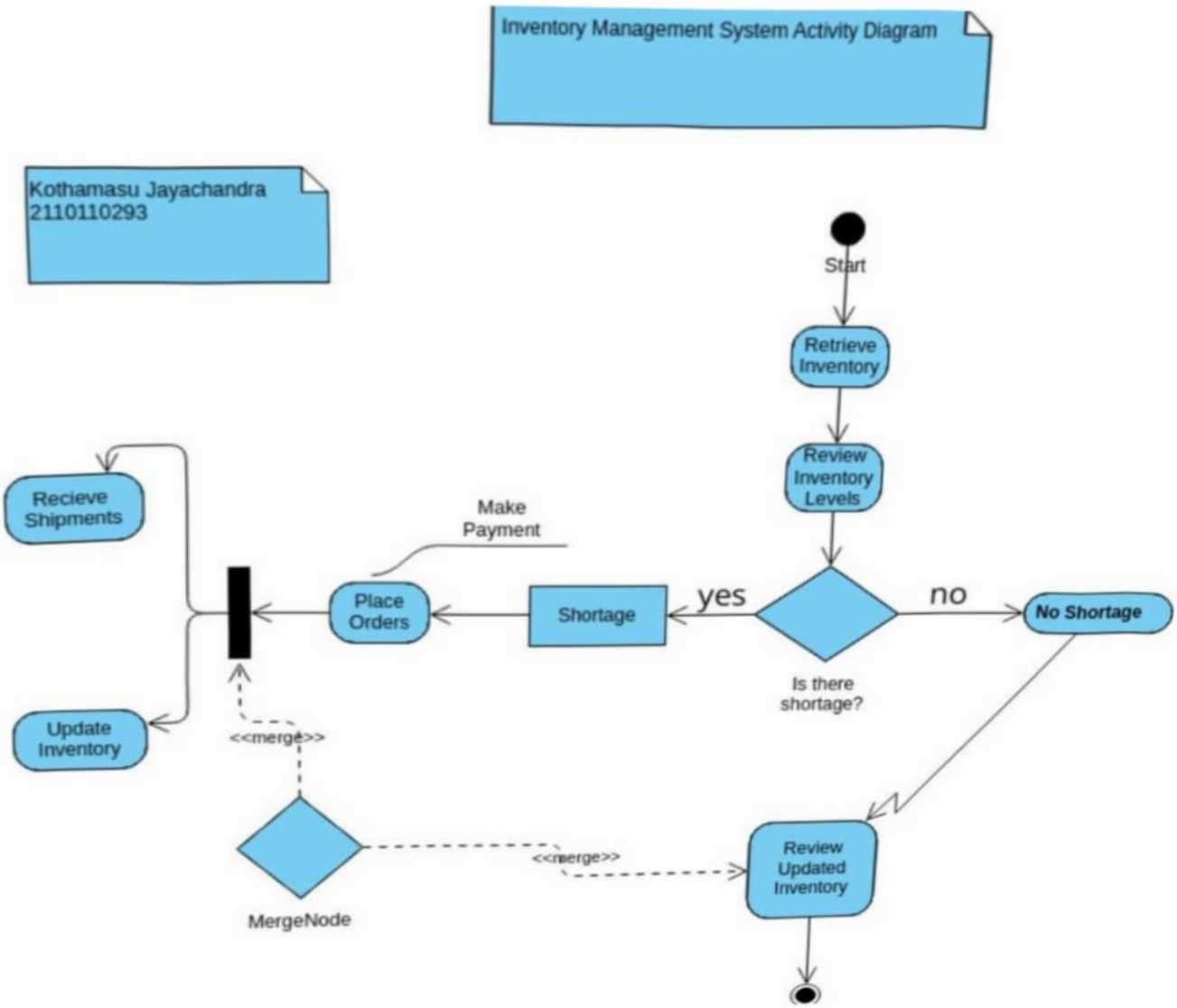
System requirements

Hardware Requirements

The hardware required to run the Product Inventory Management System includes a computer or laptop with basic specifications. The system should have at least 4GB RAM an sufficient storage space to store the database.

Software Requirements

The software required includes an operating system such as Windows or Linux. The system may be developed using programming languages such as Java, Python, or PHP. A database management system such as MySQL may be used to store product information.



User Roles

The system contains one main role which is the user.

User Responsibilities
Register in the system.

Login using email and password.

Add products to inventory.

View product list.

Update product details.

Delete products.
Search products by name

Folder Structure

The projectcontainsthefollowing folder structure:

- server
- src
- controllers
- models
- routes
- middlewares
- db

This structure keeps the project organized and easy to maintain.

Conclusion

The Product Inventory Management System provides a secure and efficient way to manage product data.

The project demonstrates the use of modern backend technologies such as Node.js, Express.js, and MongoDB.

The system supports authentication and CRUD operations for product management.